

Analysis on the vetBP Dataset

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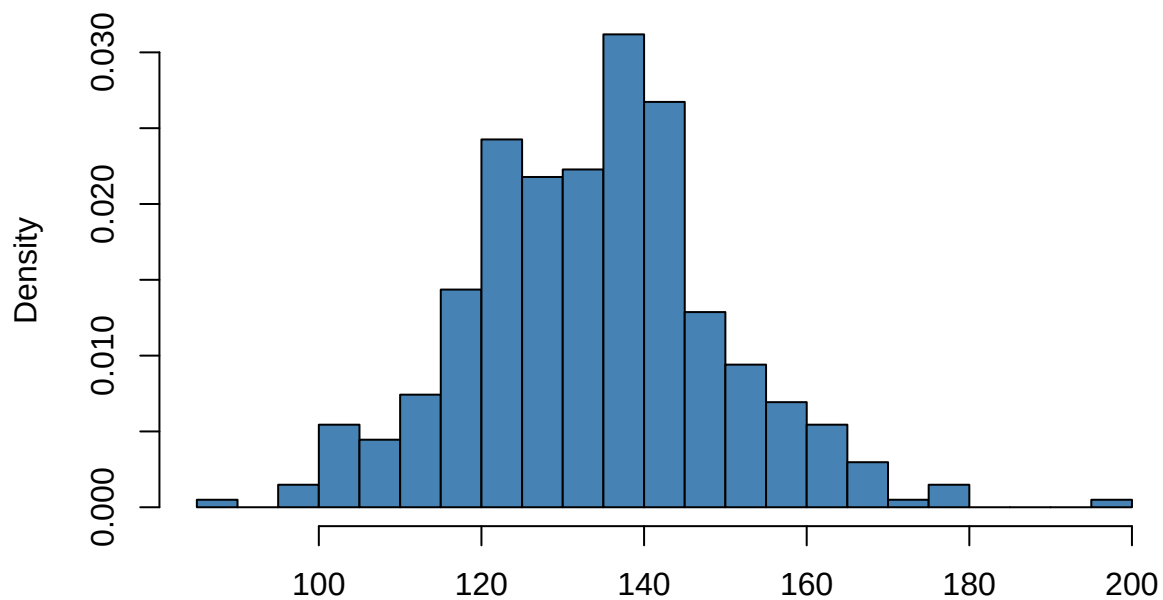
Prior elicitation on the vetBP dataset

The **vetBP** dataset collects data about blood pressures of 405 U.S. veterans. we have a measure of the systolic blood pressure at the baseline (i.e. before the treatment) and two follow-up measures (after 6 and 12 months). All systolic blood pressures are in mmHg. we have also information regarding the patient, such as its height, weight and chronic diseases.

We focus on the baseline systolic blood pressure (coded as **SBP_00**) and assume them as conditionally iid data from a Gaussian distribution.

$$y_1, \dots, y_N \mid \mu, \sigma^2 \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$$

```
sbp = df$SBP_00  
hist(sbp, probability = T, col = 'steelblue', breaks = 20, xlab = '', main = '')
```



Case 1 - Known σ^2

In this case, we assume σ^2 to be fixed (but still needs to be estimated) and we set a Normal prior for μ . Thus, we are in the case of a Normal - Normal model:

$$y_1, \dots, y_N \mid \mu \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$$
$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

Question: How can we estimate the variance σ^2 and the model parameters μ_0, σ_0^2 ?

We can not use the data on which we need to make inference! *Why?*

Thus, what it is usually done is either to have long reunions with experts gives you the prior information you need to set up the prior, or, as in our case, we look to similar data. For instance, we consider the NCHS report from 2011, in which we can find the following information:

- The average systolic pressure is (124 ± 0.3) mmHg;
- The 0.25 and 0.75 empirical percentiles of systolic pressure are, respectively, 113 mmHg and 131 mmHg.

With this information we can elicit our prior, i.e., we can set the model parameters. We do as follows:

1. From “The average systolic pressure is (124 ± 0.3) mmHg” we have information about the population mean and its standard deviation. Thus, it is reasonable to set $\mu_0 = 124$. Does it make sense to set $\sigma_0 = 0.3$?
2. From “The 0.25 and 0.75 empirical percentiles of systolic pressure are, respectively, 113 mmHg and 131 mmHg” we have information about the whole population. Hence, we can use it to estimate σ . In fact, under a Normal model, we know that $IQR = 1.34\sigma$. Thus, it is reasonable to set $\sigma = \frac{IQR}{1.34} = 13.43$.

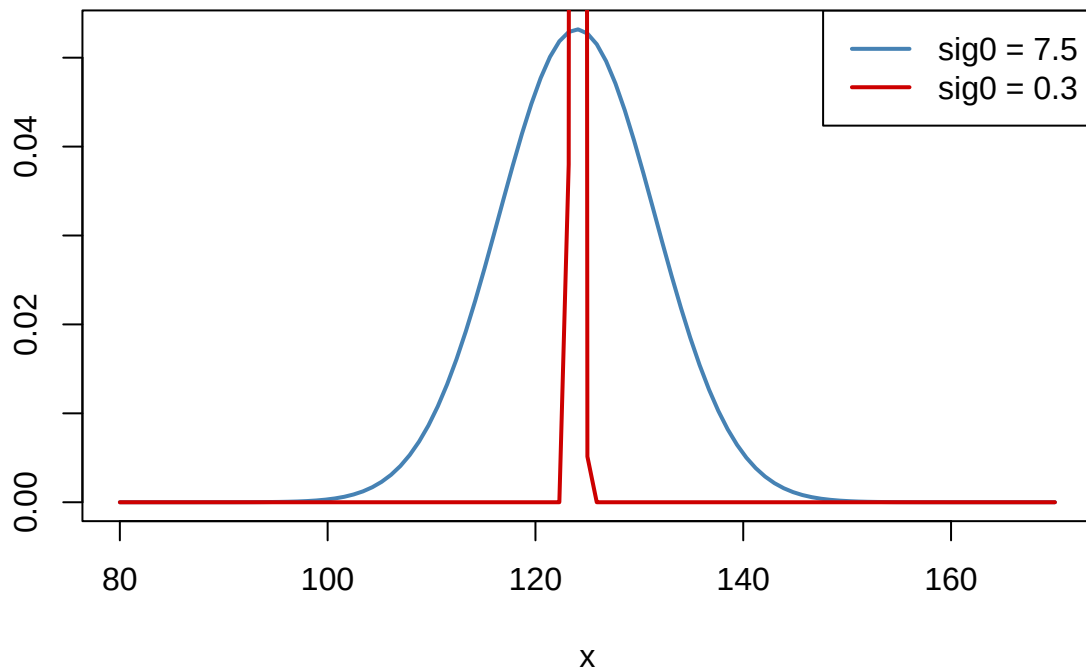
Elicited prior: $\sigma = 13.43$, $\mu_0 = 124$ and $\sigma_0 = 25 \cdot 0.3 = 7.5$.

```
# Prior parameters
mu0 = 124
sig0 = 7.5

curve(dnorm(x, mu0, sig0), xlim = c(80, 170), col = 'steelblue', ylab = '', lwd = 2)

# What happens if we use sig0 = 0.3
curve(dnorm(x, mu0, 0.3), xlim = c(80, 170), col = 'red3', ylab = '', lwd = 2, add = T)

legend(x = 'topright',
       legend = c('sig0 = 7.5', 'sig0 = 0.3'),
       col = c('steelblue', 'red3'),
       lwd = 2)
```



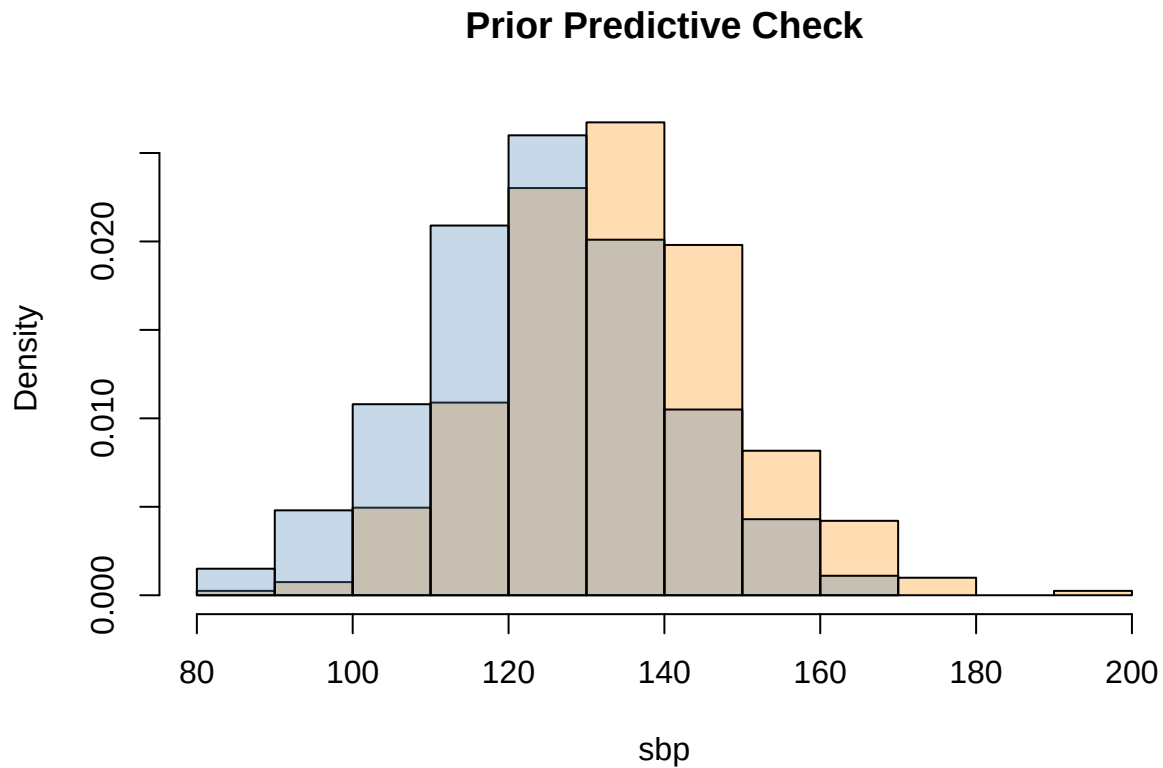
Question: How can we check if the elicited prior is *good*? A possible solution is to perform some prior predictive checks. Thus, we can compare the dataset with samples from the “prior predictive distribution”, i.e. the marginal.

```
n_samp = 1000

mu_samps <- rnorm(n_samp, mu0, sig0)
sample_from_marg <- rnorm(n_samp, mean = mu_samps, sd = 13.43)

p1 <- hist(sbp, plot = FALSE)
p2 <- hist(sample_from_marg, plot = FALSE)

plot(p1, col = rgb(1.0, 0.549, 0.0, 0.3), freq = FALSE, main = 'Prior Predictive Check')
plot(p2, col = rgb(0.275, 0.510, 0.706, 0.3), freq = FALSE, add = T)
```

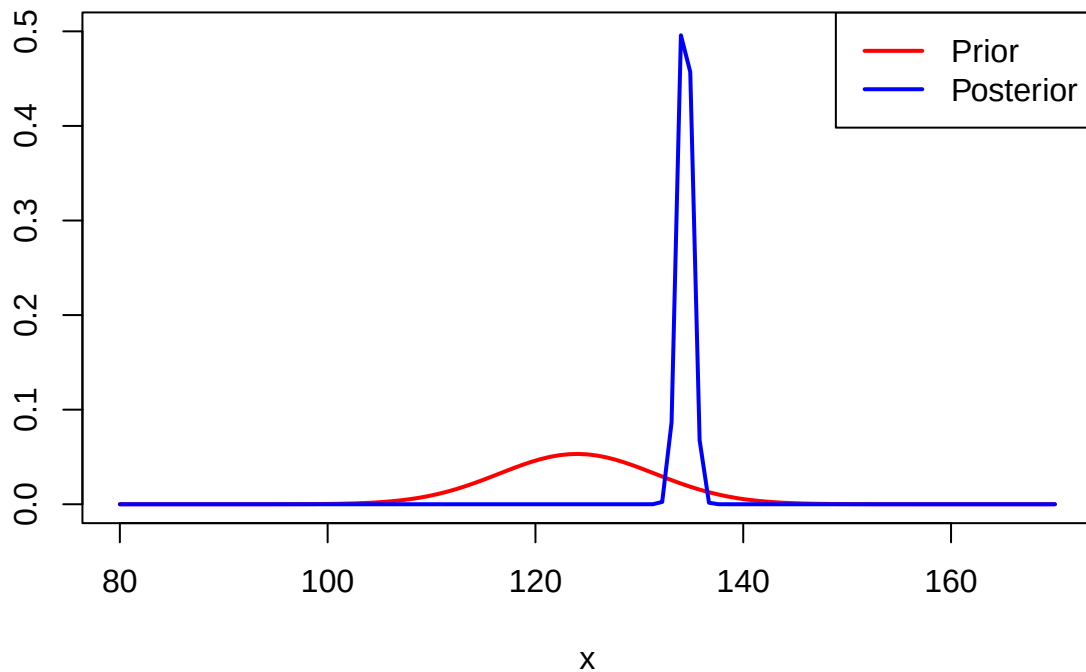


We now compute the posterior distribution. Since the model is conjugate, we simply need to update the parameters.

```
# Model precisions (1 / variance)
mu0 = 124
sig0 = 7.5
tau = 1.0 / 13.43^2
tau0 = 1.0 / sig0^2

# Compute posterior parameters
post_var1 = 1.0 / (tau * length(sbp) + tau0)
post_mean1 = (tau * sum(sbp) + tau0 * mu0) * post_var1

# Plot prior and posterior curves
# Plot
curve(dnorm(x, mu0, sig0), xlim = c(80, 170), ylim = c(0, 0.5), col = 'red', lwd = 2, ylab = '')
curve(dnorm(x, post_mean1, sqrt(post_var1)), add = T, col = 'blue', lwd = 2)
legend(
  x = "topright",
  legend = c("Prior", "Posterior"),
  col = c("red", "blue"),
  lwd = 2
)
```



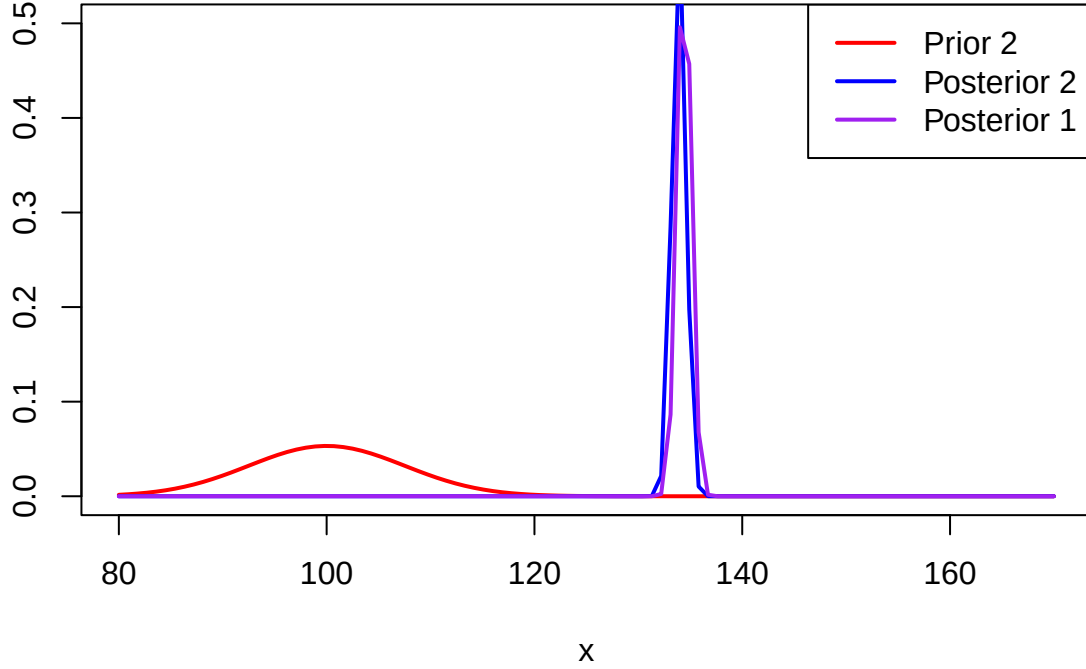
What if we set up a prior without doing prior elicitation? We may see a discrepancy between the prior and the posterior distribution. If this happens, we need to be aware that the posterior distribution may be biased by the strange prior we set.

```
# Prior parameters
tau = 1.0 / 13.43^2
tau0 = 1.0 / 5.1^2
mu0 = 100

# Posterior parameters
post_var2 = 1.0 / (tau * length(sbp) + tau0)
post_mean2 = (tau * sum(sbp) + tau0 * mu0) * post_var2

# Plot prior and posterior curves
# Plot
curve(dnorm(x, mu0, sig0), xlim = c(80, 170), ylim = c(0, 0.5), col = 'red', lwd = 2, ylab = '')
curve(dnorm(x, post_mean2, sqrt(post_var2)), add = T, col = 'blue', lwd = 2)
curve(dnorm(x, post_mean1, sqrt(post_var1)), add = T, col = 'purple', lwd = 2)

legend(
  x = "topright",
  legend = c("Prior 2", "Posterior 2", "Posterior 1"),
  col = c("red", "blue", "purple"),
  lwd = 2
)
```



Case 2 - Unknown σ^2

Now that also σ^2 is unknown, we use a Normal - Inverse Gamma model as joint prior for (μ, σ^2) . We define $\tau = 1/\sigma^2$ as the *precision*. Thus, the model can be written as:

$$\begin{aligned}
 y_1, \dots, y_N \mid \mu, \tau &\stackrel{iid}{\sim} \mathcal{N}\left(\mu; \frac{1}{\tau}\right) \\
 \mu \mid \tau &\sim \mathcal{N}\left(\mu_0, \frac{1}{\lambda\tau}\right) \\
 \tau &\sim \text{gamma}(a, b)
 \end{aligned}$$

We keep the prior elicitation as before. A good guess for the population mean is $\mu_0 \approx 124$ and a good guess for the population variance is $1/\tau \approx 13.43^2$.

We elicit our prior via plug-in estimate (also assuming something that is not true), i.e.

$$\mathbb{E}\left[\frac{1}{\tau}\right] = 13.43^2 \quad \text{but saying:} \quad \frac{1}{\mathbb{E}[\tau]} = \frac{b}{a} = 13.43^2.$$

Hence, we have set, for now, $\mu_0 = 124$ and $b = 13.43^2 \cdot a$. What about a and λ ?

```

# Compute marginal distribution for the mean (Normal-Gamma conjugate prior)
mu_marg <- function(mu, mu0, lam, a, b, log = FALSE) {
  out_log <- lgamma(a + 0.5) - lgamma(a)
  out_log <- out_log + a * log(b) - (a + 0.5) * log(b + 0.5 * lam * (mu - mu0)^2)
  out_log <- out_log + log(sqrt(lam) / sqrt(2 * pi))

  if (log) {
    return(out_log)
  } else {
    return(exp(out_log))
  }
}

mu0 <- 124
mugrid <- seq(100, 200, length.out = 1000)
a_vals <- c(0.5, 1, 2.0, 5)
lam_vals <- c(0.01, 0.05, 0.1, 0.5, 1.0, 5.0)

# Setup 1x4 subplot grid
par(mfrow = c(1, length(a_vals)), mar = c(4, 4, 3, 1))

# Palette for lambda lines
colors <- rainbow(n = length(lam_vals), end = 0.8)

for (i in seq_along(a_vals)) {
  a <- a_vals[i]
  b <- 13.43^2 * a

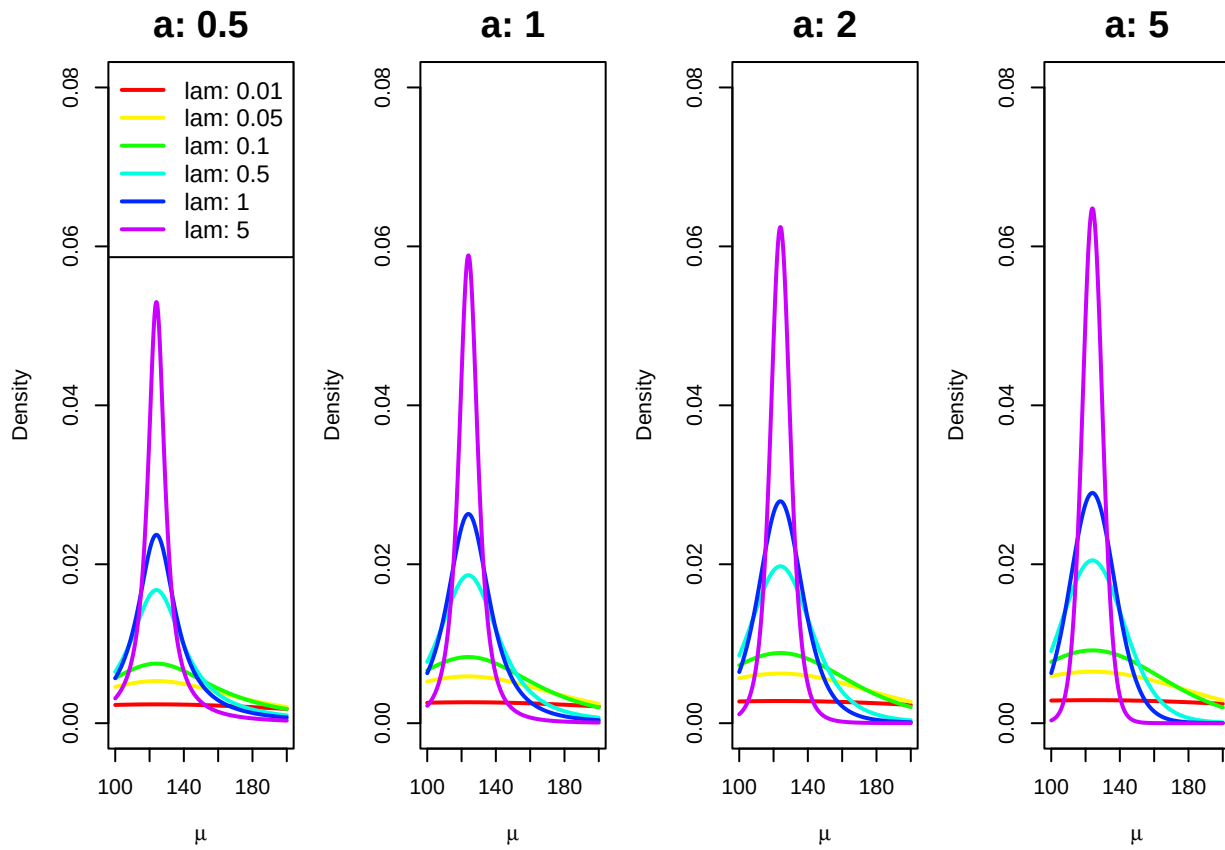
  # Compute density for first lambda to initialize plot frame
  first_dens <- mu_marg(mugrid, mu0, lam_vals[1], a, b)

  plot(
    mugrid, first_dens, type = "l", col = colors[1], lwd = 2,
    ylim = c(0, 0.08), xlab = expression(mu), ylab = "Density",
    main = paste0("a: ", a), cex.main = 1.8
  )

  # Add remaining lambda lines
  for (j in 2:length(lam_vals)) {
    dens <- mu_marg(mugrid, mu0, lam_vals[j], a, b)
    lines(mugrid, dens, col = colors[j], lwd = 2)
  }

  if (i == 1) {
    legend("topright", legend = paste0("lam: ", lam_vals), col = colors, lwd = 2, cex = 1.1)
  }
}

```



```
# Reset par settings
par(mfrow = c(1, 1))
```

```
lam_vals <- 5
```

```
# Palette for lambda lines
colors <- rainbow(n = length(a_vals), end = 0.8)
a_first <- a_vals[1]
b_first <- 13.43^2 * a_first
first_dens <- mu_marg(mugrid, mu0, 5, a_first, b_first)

plot(
  mugrid, first_dens, type = "l", col = colors[1], lwd = 2,
  ylim = c(0, 0.08), xlab = expression(mu), ylab = "Density",
  main = expression(paste("Marginal Density of ", mu, " for different values of ", a))
)

if (length(a_vals) > 1) {
  for (i in 2:length(a_vals)) {
    a <- a_vals[i]
    b <- 13.43^2 * a
    dens <- mu_marg(mugrid, mu0, 5, a, b)

    lines(mugrid, dens, col = colors[i], lwd = 2)
  }
}
```

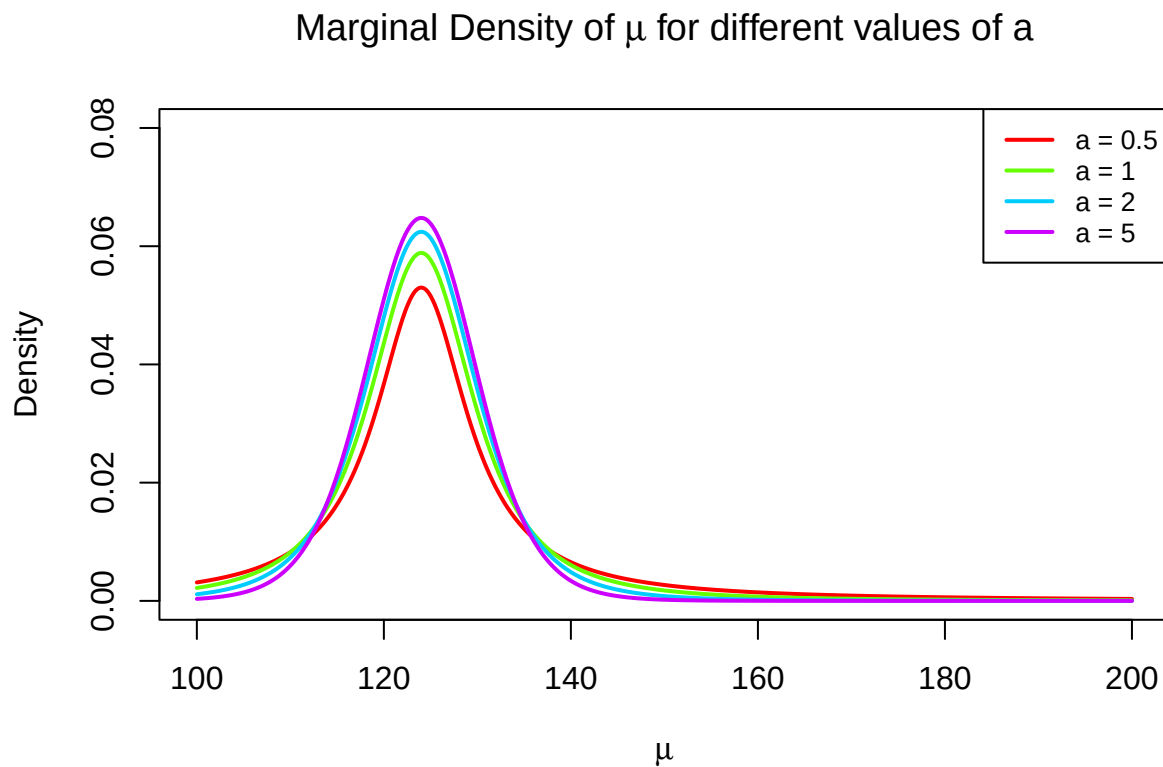


```

}

legend(
  "topright",
  legend = paste0("a = ", a_vals),
  col = colors,
  lwd = 2,
  cex = 0.8
)

```



Elicited Prior: $\lambda = 5$, $a = 2$, $\mu_0 = 124$ and $b = a \cdot 13.43^2$

Let's look at the posterior distribution

```

# Compute posterior parameters for Normal-Gamma conjugate model
get_post_params <- function(y, mu0, lam, a, b) {
  ybar <- mean(y)
  n <- length(y)
  s <- mean((y - ybar)^2)

  apost <- a + n / 2.0
  bpost <- b + 0.5 * (n * s + (lam * n * (ybar - mu0)^2) / (lam + n))
  lampost <- lam + n
  mupost <- (lam * mu0 + n * ybar) / (lam + n)
}

```

```

return(list(
  mupost = mupost,
  lampost = lampost,
  apost = apost,
  bpost = bpost
))
}

```

```

# Prior parameters
mu0 <- 124
lam <- 5
a <- 2
b <- a * 13.43^2

# Posterior parameters
post <- get_post_params(sbp, mu0, lam, a, b)

mupost <- post$mupost
lampost <- post$lampost
apost <- post$apost
bpost <- post$bpost

```

```

# Setup 1x2 layout
par(mfrow = c(1, 2), mar = c(4.5, 4.5, 3, 1))
mugrid <- seq(100, 160, length.out = 1000)
taugrid <- seq(0, 0.01, length.out = 1000)

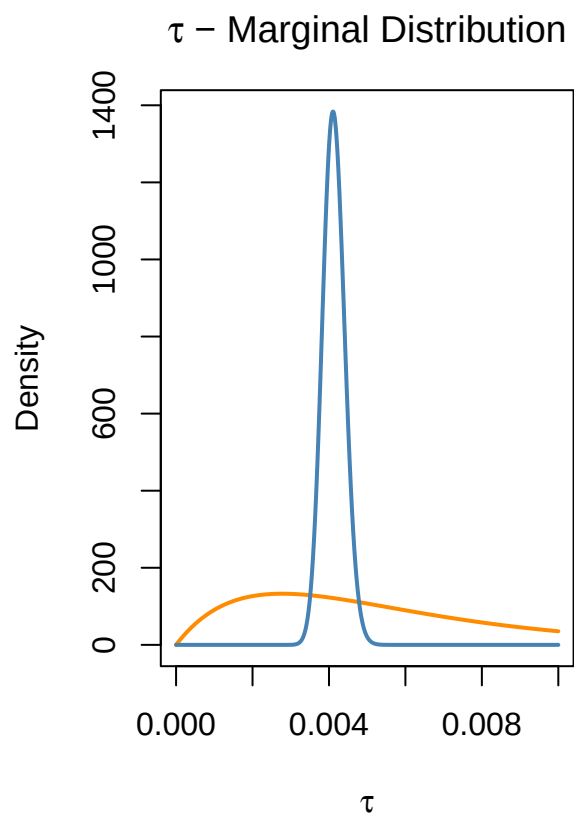
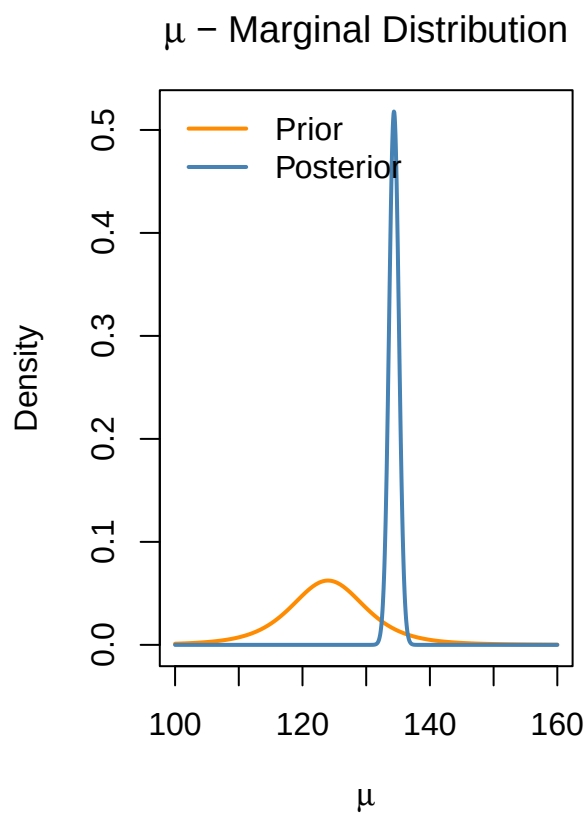
# Plot mu
mu_prior_dens <- mu_marg(mugrid, mu0, lam, a, b)
mu_post_dens <- mu_marg(mugrid, mupost, lampost, apost, bpost)

plot(mugrid, mu_prior_dens, type = "l", col = "darkorange", lwd = 2,
     xlab = expression(mu), ylab = "Density", main = expression(mu * " - Marginal Distribution"),
     ylim = range(c(mu_prior_dens, mu_post_dens)), cex.main = 1.2)
lines(mugrid, mu_post_dens, col = "steelblue", lwd = 2)
legend("topleft", legend = c("Prior", "Posterior"), col = c("darkorange", "steelblue"), lwd = 2, bty = "n")

# Plot tau
tau_prior_dens <- dgamma(taugrid, shape = a, rate = b)
tau_post_dens <- dgamma(taugrid, shape = apost, rate = bpost)

plot(taugrid, tau_prior_dens, type = "l", col = "darkorange", lwd = 2,
     xlab = expression(tau), ylab = "Density", main = expression(tau * " - Marginal Distribution"),
     ylim = range(c(tau_prior_dens, tau_post_dens)), cex.main = 1.2)
lines(taugrid, tau_post_dens, col = "steelblue", lwd = 2)

```



```
# Reset graphics layout
par(mfrow = c(1, 1))
```

```
apost / bpost
```

```
## [1] 0.004125956
```